

Smaller notes

Matthew Daws

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1 Unions of cosets

This is a careful write-up of my question [4] at *mathoverflow* together with the answer [13]. The question is:

Let G be a locally compact group, and let K, L be cosets of G (not assumed open or closed) which each have empty interior. Does also $K \cup L$ have empty interior?

The answer is “no”. The counter-example comes from considering $G = (\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$ the infinite product of the cyclic group of order 2. We shall write 0, 1 for the elements of $\mathbb{Z}/2\mathbb{Z}$, so $1 + 1 = 0$. We give G the product topology, so G is compact and Hausdorff. We shall write elements of G as infinite sequences $x = (x_n)$ in $\mathbb{Z}/2\mathbb{Z}$. Notice that G is compact, abelian, and every $x \in G$ satisfies that $x + x = 0$.

The topology has a base of “cylinder” sets, given as follows. Let $n \in \mathbb{N}$ and $y = (y_1, y_2, \dots, y_n)$ a finite sequence in $\mathbb{Z}/2\mathbb{Z}$. Define

$$\mathcal{O}_{n,y} = \{x = (x_k) \in G : x_i = y_i \ (i \leq n)\}.$$

These sets form a base for the topology on G . Notice that the intersection of two such sets is of the same form (or is empty).

Furthermore, notice that for $s, t \in \mathbb{Z}/2\mathbb{Z}$ either $s = t$ or $s = t + 1$. Then

$$\begin{aligned} G \setminus \mathcal{O}_{n,y} &= \{x : \exists 1 \leq i \leq n, x_i \neq y_i\} = \bigcup_{i=1}^n \{x : x_i \neq y_i\} \\ &= \bigcup_{i=1}^n \{x : x_i = y_i + 1\} \end{aligned}$$

which is the union of (many) basic open sets. Thus $\mathcal{O}_{n,y}$ is also closed.

Finally, notice that $\mathcal{O}_{n,y}$ is a subgroup exactly when $y_i = 0$ for $i \leq n$, and so every $\mathcal{O}_{n,y}$ is a coset in G .

Lemma 1.1. *G has countably many open subgroups.*

Proof. Consider a basic open set $\mathcal{O}_{n,y}$. Given $x, z \in \mathcal{O}_{n,y}$, as x and z agree in the first n coordinates, we see that $(x + z)_k = 0$ for $k \leq n$. It follows that $\mathcal{O}_{n,y} + \mathcal{O}_{n,y} = \mathcal{O}_{n,0}$ an open subgroup.

Now let H be an arbitrary open subgroup, so we can write H as some union of basic open sets. Let n be minimal with $\mathcal{O}_{n,y} \subseteq H$ for some y . Thus $\mathcal{O}_{n,0} \subseteq H$ as H is a subgroup. Any open basic open set $\mathcal{O}_{m,z} \subseteq H$ must have $n \leq m$, and so we see that $\mathcal{O}_{m,z} + \mathcal{O}_{n,0} = \mathcal{O}_{n,z} \subseteq H$.

As H is the union of basic open sets, we conclude that there are finitely many $x_1, \dots, x_k$ with $H = \bigcup_{i=1}^k \mathcal{O}_{n,x_i}$. Let $y_i \in (\mathbb{Z}/2\mathbb{Z})^n$ be the projection of x_i onto the first n coordinates. As H is a subgroup, it follows that $\{y_i : 1 \leq i \leq k\}$ is a subgroup of $(\mathbb{Z}/2\mathbb{Z})^n$, say K . Furthermore, H is exactly the collection of all $x \in G$ such that the projection of x onto the first n coordinates is in K .

It follows that open subgroups of H can be described by $n \in \mathbb{N}$ and a subgroup K of $(\mathbb{Z}/2\mathbb{Z})^n$. There are only countably many such choices. $\square$

Corollary 1.2. *There are countably many closed subgroups of G of index 2.*

Proof. Let $H \leq G$ be a closed subgroup of index 2. Then $G \setminus H$ is a coset of H and so is closed, and so H is open. The result follows. $\square$

We now consider arbitrary subgroups of G . It is instructive to consider the bijection between G and $\mathcal{P}(\mathbb{N})$ the power set of $\mathbb{N}$, given by $x = (x_n)$ mapping to the set $A \subseteq \mathbb{N}$ where $n \in A$ if and only if $x_n = 0$. If $x, y \in G$ biject with A, B , respectively, then $x + y$ bijects with C where $n \in C$ if and only if $x_n + y_n = 0$, that is, $x_n = y_n = 0$ or $x_n = y_n = 1$, that is, $n \in A \cap B$ or $n \in \mathbb{N} \setminus (A \cup B)$. Thus $C = (A \cap B) \cup (\mathbb{N} \setminus (A \cup B)) = \mathbb{N} \setminus (A \Delta B)$.

We recall the notion of a *filter* on $\mathbb{N}$. This is a subset $\mathcal{F} \subseteq \mathcal{P}(\mathbb{N})$ with $\emptyset \notin \mathcal{F}$, with, if $A \in \mathcal{F}$ and $A \subseteq B$ then also $B \in \mathcal{F}$, and with $A, B \in \mathcal{F} \implies A \cap B \in \mathcal{F}$. An *ultrafilter* $\mathcal{U}$ is a maximal filter; alternatively, $\mathcal{U}$ is a filter with the property that if $A \in \mathcal{P}(\mathbb{N})$ then either $A \in \mathcal{U}$ or $\mathbb{N} \setminus A \in \mathcal{U}$.

Lemma 1.3. *There are 2^c subgroups of index 2 in G .*

Proof. Let $\mathcal{U}$ be an ultrafilter, and let $H \subseteq G$ be the associated subset. Given $A, B \in \mathcal{U}$ consider $C = \mathbb{N} \setminus (A \Delta B)$. Then $A \cap B \subseteq C$ and so $C \in \mathcal{U}$. It follows that H is a subgroup of G . Furthermore, given $A \notin \mathcal{U}$ we know that $\mathbb{N} \setminus A \in \mathcal{U}$. Thus if $x \notin H$ then $1 + x \in H$, and as $\emptyset \notin \mathcal{U}$ also $1 \notin H$. Thus H is proper, and G is the union of H and $1 + H$, so H has index 2.

It is well-known (see for example [10]) that there are 2^c ultrafilters on $\mathbb{N}$, and so there are (at least) 2^c subgroups of index 2 in G . As G bijects with $\mathcal{P}(\mathbb{N})$ we have $|G| = 2^{\aleph_0} = \mathfrak{c}$ and so $|\mathcal{P}(G)| = 2^c$. Thus there are at most 2^c subgroups of any index. $\square$

There hence exists a subgroup H of index 2 which is not closed. (In fact this follows more directly from the existence of non-principle ultrafilters, and the proof of Lemma 1.1.) Thus H is not open, and so cannot contain any non-empty open set (if $\emptyset \neq U \subseteq H$ is open then using the group operations we can cover H by translates of U which shows that H is open, contradiction). As $G \setminus H$ is a coset of H it follows that $G \setminus H$ cannot contain any non-empty open set. Thus H is dense in G . We have also now answered our original question, as both H and its coset have empty interior, and yet their union is all of G .

2 Semi-direct products

This is standard material. Let G be a group with a subgroup H and a normal subgroup N . The following statements are equivalent:

1. $G = NH = \{nh : n \in N, h \in H\}$ and $N \cap H = \{e\}$;
2. for each $g \in G$ there are unique $n \in N, h \in H$ with $g = nh$;
3. for each $g \in G$ there are unique $n \in N, h \in H$ with $g = hn$;
4. for the inclusion $i : H \rightarrow G$ and the quotient $\pi : G \rightarrow G/N$, the composition $\pi \circ i : H \rightarrow G/N$ is an isomorphism;
5. there is a homomorphism $G \rightarrow H$ that is the identity on H and has kernel N ;
6. there is a split short-exact sequence $1 \rightarrow N \rightarrow G \rightarrow H \rightarrow 1$.

Let us show these equivalences. If (1) holds then $G = NH$ but if there was a possibly non-unique way to write $g = n_1 h_1 = n_2 h_2$ then $n_2^{-1} n_1 = h_2 h_1^{-1} \in N \cap H = \{e\}$ so $n_1 = n_2$ and $h_1 = h_2$, so (2) holds. Conversely, the uniqueness clause in (2) shows that if $g \in N \cap H$ then we must have $g = e$. So (1) $\Leftrightarrow$ (2).

That (1) $\Leftrightarrow$ (3) is similar, using the group inverse to show that $G = NH$ if and only if $G = G^{-1} = H^{-1}N^{-1} = HN$.

Considering (4), $\ker(\pi \circ i) = N \cap H$ and the image of $\pi \circ i$ is $\{Nh : h \in H\} = \{Nnh : n \in N, h \in H\}$, so immediately (1) implies (4), while conversely, if $\pi \circ i$ is onto, then given $g \in G$ there is $h \in H$ with $Nh = Ng$ so $g \in Nh \subseteq NH$ and hence (1) holds. So (1) $\Leftrightarrow$ (4).

For (5) consider $p : G \rightarrow H$ a homomorphism with $p(h) = h$ for $h \in H$ and $\ker(p) = N$. Then $N \cap H = \{e\}$ while for $g \in G$ let $h = p(g) \in H$ so $p(hg^{-1}) = hp(g)^{-1} = e$ and hence $hg^{-1} = n$ for some $n \in N$, so $g = n^{-1}h \in NH$, so (1) holds. Conversely, given (1), also (4), let $p : G \rightarrow H$ be the composition of $(\pi \circ i)^{-1} \circ \pi : G \rightarrow H$. Then p is a homomorphism, $\ker(p) = \ker(\pi) = N$, and for $h \in H$ we have $p(h) = h_1$ where $h_1 \in H$ is the necessarily unique element with $h_1 N = \pi(h) = hN$, so $h_1 = h$ by uniqueness. Hence (5) holds.

For (6) we have

$$1 \longrightarrow N \xhookrightarrow{\iota} G \xrightarrow[\theta]{\pi} H \longrightarrow 1 \quad (1)$$

where ι is an inclusion, π a surjection with $\ker(\pi) = \iota(H)$, and we have $\pi \circ \theta = \text{id}_H$. We identify N with its image in G which is certainly normal, and then π gives an isomorphism between G/N and H . That $\pi \circ \theta = \text{id}_H$ means θ is injective, so we identify H with its image in G . Then $p = \theta \circ \pi$ is an idempotent homomorphism, which is the identity on H , and has $\ker(p) = \ker(\pi) = N$, so (5) holds. Conversely, given (5), we have inclusions $N \rightarrow G$ and $H \rightarrow G$ giving ι and θ . Consider the map $p : G \rightarrow H$, and set $\pi = p$. Then $\ker(\pi) = \ker(p) = N$ while π is onto, and $\pi \circ \theta = \text{id}_H$, so (6) holds.

Any of these equivalent conditions define what it means for G to be the (*inner*) *semidirect product* denoted $G = N \rtimes H$ (sometimes $G = H \ltimes N$).

Definition 2.1. The (*outer*) *semidirect product* of groups N, H is defined by specifying a homomorphism $\varphi : H \rightarrow \text{Aut}(N)$ and setting $G = N \rtimes H$ as a set, with product

$$(n_1, h_1)(n_2, h_2) = (n_1 \varphi(h_1)(n_2), h_1 h_2).$$

The identity of G is (e_N, e_H) and the inverse is $(n, h)^{-1} = (\varphi(h^{-1})(n^{-1}), h^{-1})$.

Given such a construction, we identify N with $\{(n, e_H) : n \in N\}$ and H with $\{(e_N, h) : h \in H\}$. These are seen to be subgroups of G . Then $NH = \{(n, e)(e, h) : n \in N, h \in H\} = G$ and $N \cap H = \{e\}$. Finally, the product satisfies

$$hnh^{-1} = (e, h)(n, e)(e, h)^{-1} = (\varphi(h)(n), h)(e, h^{-1}) = (\varphi(h)(n)\varphi(h)(e), e) = \varphi(h)(n) \in N,$$

for each $n \in N, h \in H$. Thus N is a normal subgroup and we have verified condition (1).

Conversely, if we have an inner semidirect product $G = NH$ with $N \trianglelefteq G$, then define $\varphi(h)(n) = hnh^{-1} \in N$, so that $\varphi(h)$ is an automorphism of N for each h , and $\varphi : H \rightarrow \text{Aut}(N)$ is an homomorphism. We define $\theta : G \rightarrow N \times H$ by $\theta(nh) = (n, h)$. As a slight aside, notice that this map is a homomorphism if and only if $\theta(h)\theta(n) = \theta(hn) = \theta((hnh^{-1})h) = \theta(\varphi(h)(n)h)$, which is true by definition. This “commutation relation”, showing how elements of N and H commute pass each other, is often a useful way to think of a semi-direct product.

Thus inner and outer semidirect products are canonically isomorphic.

Finally, we claim that G is isomorphic to $N \rtimes H$ if and only if there is a split short exact sequence

$$1 \longrightarrow N \xhookrightarrow{\beta} G \xrightarrow[\gamma]{\alpha} H \longrightarrow 1$$

This is just condition (6). Notice

$$\varphi(h)(n) = hnh^{-1} = \beta^{-1}(\gamma(h)\beta(n)\gamma(h^{-1}))$$

gives the action of H on N .

2.1 For topological groups

When G is a topological group, it seems natural to consider the continuous automorphisms of N , and to ask for $\varphi : H \rightarrow \text{Aut}(N)$ to be continuous. Forming the (outer) semidirect product, we form the topological product $N \times H$ and the product as above. We require that $H \times N \rightarrow N; (h, n) \mapsto \varphi(h)(n)$ is (jointly) continuous, and then we see that the product on $N \rtimes H$ is (jointly) continuous. The inverse is continuous, as it is the composition of $(h, n) \mapsto (h^{-1}, n^{-1})$ with the continuous action map.

Then N, H are closed subgroups of $N \rtimes H$. Conversely, start with $G = NH$ for some closed N, H with $N \trianglelefteq G$. Then in (4) while i, π are continuous, so $\pi \circ i$ is continuous, we require that $(\pi \circ i)^{-1}$ be continuous as well. This is equivalent to p in (5) being continuous. For (6), in the split short exact sequence (1), we have that ι, π are continuous, and we also require that θ is continuous. Then θ is injective, and furthermore is a homeomorphism onto its range, because $\pi \circ \theta = \text{id}_H$.

It does not seem obvious what conditions we would like to add to (1), (2) or (3) to ensure the correct continuity conditions. However, notice that given $G = NH$ with $N \trianglelefteq H$, with N, H closed, we can still define $\varphi(h)(n) = hnh^{-1}$, and continuity of the product in G will ensure that $\varphi(h)$ is continuous for h , and that $H \times N \rightarrow N; (h, n) \mapsto \varphi(h)(n)$ is continuous. Thus we can form the outer semidirect product. The canonical isomorphism is $G \rightarrow N \rtimes H, nh \mapsto (n, h)$. The inverse of this is always continuous, but this map itself might not be. It is continuous exactly when $(n_i h_i)$ is a net converging to e , we must have $n_i \rightarrow e$ and $h_i \rightarrow e$. If G is locally compact, this fails to happen exactly when we can find $(n_i) \subseteq N, (h_i) \subseteq H$ with $n_i \rightarrow \infty, h_i \rightarrow \infty$ and yet $n_i h_i \rightarrow e$.

Remark 2.2. It is implicitly claimed on page 10 of [8] that in this setting, the map from the outer semidirect product $N \rtimes H$ to G is an isomorphism of locally compact groups. The following counter-example, from [14], shows that this need not be the case. Let K be an infinite compact group and let K_d be K with the discrete topology, so $K \neq K_d$ as topological groups.

Let $G = K_d \times K$ the direct product of groups, so G is locally compact. Let $N = K_d \times \{e\}$ a closed subgroup, and let $H = \{(g, g) : g \in K\} \subseteq G$ the diagonal, which is also a closed, normal subgroup. Then it's easy to see that $N \cap H = \{e\}$ and $NH = G$. However, both N and H have the discrete topology (as the subspace topologies from G) and so the outer semidirect product $N \rtimes H$ will also be discrete. Hence $N \rtimes H \rightarrow G$ is continuous, but the inverse is not.

In this example, the map p from (5) is given by $p(g, h) = (h, h)$ which is obviously a projection onto the diagonal H with kernel N . This is not continuous, as there is a net (h_i) in K which converges to e without eventually being equal to e . Then for any fixed g , we have $(g, h_i) \rightarrow (g, e)$ in G but $(h_i, h_i) \not\rightarrow (e, e)$. A similar remark applies to (6), while in (4) we see that $\pi \circ i$ does not have continuous inverse.

In [8], see page 9, in the locally compact case the Haar measure on $N \rtimes H$ is computed. For each $h \in H$ the measure on N given by $\lambda_N^h(E) = \lambda_N(\varphi(h)(E))$, for each Borel E , is left-invariant, and so there is $\delta(h) > 0$ with $\lambda_N^h(E) = \delta(h)\lambda_N(E)$ for all E . One can check that $\delta : H \rightarrow (\mathbb{R}^+, \times)$ is a continuous homomorphism, and then a left Haar measure on $N \rtimes H$ is given by

$$\int_{N \rtimes H} f(n, h) d(n, h) = \int_H \int_N f(n, h) \delta(h)^{-1} dn dh. \quad (f \in C_{00}(N \rtimes H)).$$

The modular function is

$$\Delta_{N \rtimes H}(n, h) = \Delta_N(n) \Delta_H(h) \delta(h)^{-1}.$$

3 Modules over algebras of operators

The following is surely folklore: people will know the construction, but I'm not aware of a reference.

We consider homomorphisms $\pi: \mathcal{B}(E) \rightarrow \mathcal{B}(X)$ for Banach spaces E and X . Any such π restricts to a homomorphism $\pi: \mathcal{F}(E) \rightarrow \mathcal{B}(X)$, where $\mathcal{F}(E)$ is the finite-rank operators. For $e \in E, e^* \in E^*$ we write $e \otimes e^*$ for the rank-one operator $E \ni f \mapsto \langle e^*, f \rangle e$.

Choose $e_0 \in E, e_0^* \in E^*$ with $\langle e_0^*, e_0 \rangle = 1$, so that $e_0 \otimes e_0^* \in \mathcal{F}(E)$ is an idempotent, and hence so too is $\pi(e_0 \otimes e_0^*)$. Set

$$Y_0 = \{x \in X : \pi(e_0 \otimes e_0^*)x = x\},$$

a closed subspace (as the range of an idempotent) in X . Define $\phi: E \otimes Y_0 \rightarrow X$ by $e \otimes y \mapsto \pi(e \otimes e_0^*)y$.

Lemma 3.1. *ϕ is an injective map, the image of which is a $\mathcal{B}(E)$ -submodule of X . Infact, the image of ϕ is $\text{lin}\{\pi(T)x : x \in X, T \in \mathcal{F}(E)\}$.*

Proof. Given $u \in E \otimes Y_0$ we can find linearly independent $e_1, \dots, e_n$ in E with $u = \sum_{i=1}^n e_i \otimes y_i$ for some y_i in Y . Then there are $e_j^* \in E^*$ with $\langle e_j^*, e_i \rangle = \delta_{i,j}$. Suppose $\phi(u) = 0$ we see that

$$0 = \pi(e_0 \otimes e_j^*) \sum_{i=1}^n \phi(e_i \otimes y_i) = \sum_{i=1}^n \pi(e_0 \otimes e_j^*) \pi(e_i \otimes e_0^*) y_i = \pi(e_0 \otimes e_0^*) y_j = y_j,$$

as $y_j \in Y_0$, and using that π is a homomorphism. Hence $y_j = 0$ for all j and so $u = 0$, showing that ϕ is injective.

Given $T \in \mathcal{B}(E)$ and $\phi(e \otimes y)$ we see that $\pi(T)\phi(e \otimes y) = \pi(T)\pi(e \otimes e_0^*)y = \pi(T(e) \otimes e_0^*)y = \phi(T(e) \otimes y)$, and so the image of ϕ is a $\mathcal{B}(E)$ -submodule.

The space which we claim equals the image of ϕ is $\text{lin}\{\pi(e \otimes e^*)x : x \in X, e \in E, e^* \in E^*\}$. Clearly the image of ϕ is contained in this. Conversely, given $x' = \pi(e \otimes e^*)x$, set $y = \pi(e_0 \otimes f^*)x'$ where $f^* \in E^*$ is chosen so that $\langle f^*, e \rangle = 1$ (such a functional exists via Hahn–Banach). Then $y = \pi(e_0 \otimes e^*)x$ and so $\pi(e_0 \otimes e_0^*)y = y$, hence $y \in Y_0$. Then $\phi(e \otimes y) = \pi(e \otimes e_0^*)y = \pi(e \otimes e_0^*)\pi(e_0 \otimes e^*)x = \pi(e \otimes e^*)x = x'$ and so x' is in the image of ϕ , as required. $\square$

In particular, the image of ϕ does not depend upon the choice of e_0, e_0^* . Let's explore this a little more: suppose we also have $\langle e_1^*, e_1 \rangle = 1$ and analogously define Y_1 and ϕ_1 . Define $\alpha_{1,0}: Y_0 \rightarrow Y_1$ by $y \mapsto \pi(e_1 \otimes e_0^*)y$. It's easy to see that $\alpha_{1,0}$ does map into Y_1 . Analogously define $\alpha_{0,1}$. For $y \in Y_0$ we see that

$$\alpha_{0,1}\alpha_{1,0}y = \pi(e_0 \otimes e_1^*)\pi(e_1 \otimes e_0^*)y = \pi(e_0 \otimes e_0^*)y = y,$$

and similarly, $\alpha_{1,0}\alpha_{0,1}$ is the identity on Y_1 . Furthermore, for $y \in Y_0$,

$$\phi_1(e \otimes \alpha_{1,0}y) = \pi(e \otimes e_1^*)\alpha_{1,0}y = \pi(e \otimes e_1^*)\pi(e_1 \otimes e_0^*)y = \pi(e \otimes e_0^*)y = \phi(e \otimes y).$$

So we have explicitly found intertwiners between the different maps.

Of course, this construction only tells us about homomorphisms $\mathcal{F}(E) \rightarrow \mathcal{B}(X)$. If we have a homomorphism of the generalised Calkin algebra $\mathcal{B}(E)/\mathcal{K}(E)$ then we learn nothing.

When $\mathcal{B}(X)$ is an essential $\mathcal{A}(E)$ -module, the map ϕ realises X as the completion of $E \otimes Y_0$. The action of $\mathcal{A}(E)$ (or $\mathcal{B}(E)$) is just the standard action on the E tensor factor.

If $S \in \mathcal{B}(X)$ commutes with each $\pi(T)$, for $T \in \mathcal{A}(E)$, then $y \in Y_0$ implies $Sy \in Y_0$ and then $\phi(e \otimes Sy) = S\phi(e \otimes y)$. Letting S_0 be the restriction of S to Y_0 , if we view X as the completion of $E \otimes Y_0$, then $1 \otimes S_0$ is the operator S . However, in general there seems no reason to suspect that every member of $\mathcal{B}(Y_0)$ occurs as some S_0 .

4 Polar decompositions

This material can be found in a variety of sources, but I wanted something in my own presentation to reference. Fix Hilbert spaces H, K .

Let $T \in \mathcal{B}(H, K)$ and use the continuous functional calculus to define $|T| = (T^*T)^{1/2} \in \mathcal{B}(H)$. This is the unique positive operator S with $S^2 = T^*T$. We start with some elementary facts about operators on Hilbert spaces.

Lemma 4.1. *Let $S \in \mathcal{B}(H, K)$. Then:*

- (1) $\text{Im}(S)^\perp = \ker S^*$ and so $\overline{\text{Im}}(S) = (\ker S^*)^\perp$.
- (2) $\ker S = \ker S^*S = \ker |S|$.
- (3) S is an isometry if and only if $S^*S = 1$.

Proof. We see that $\xi \in \text{Im}(S)^\perp$ if and only if $(\xi|S\eta) = 0$ for all η , if and only if $S^*\xi = 0$, so (1) follows. For (2) we note that

$$S\xi = 0 \implies S^*S\xi = 0 \implies (\xi|S^*S\xi) = 0 \implies \|S\xi\|^2 = 0 \implies S\xi = 0,$$

and so we have equivalence throughout. As $|S|^*|S| = S^*S$ also $\ker |S| = \ker S^*S$.

For (3), if $S^*S = 1$ then $\|S\xi\|^2 = (\xi|S^*S\xi) = \|\xi\|^2$ for each $\xi \in H$ and so S is an isometry. For the converse, we use the polarisation identity, so for $\xi, \eta \in H$,

$$(S\xi|S\eta) = \frac{1}{4} \sum_{k=0}^3 i^k \|S\xi + (-i)^k S\eta\|^2 = \frac{1}{4} \sum_{k=0}^3 i^k \|\xi + (-i)^k \eta\|^2 = (\xi|\eta),$$

in the middle step using that S is an isometry. It follows that $S^*S = 1$. □

Next we look at partial isomerisms in the abstract.

Lemma 4.2. *Let A be a C^* -algebra and let $u \in A$. The following are equivalent:*

- (1) u^*u is a projection;
- (2) $uu^*u = u$;
- (3) $u^*uu^* = u^*$;
- (4) uu^* is a projection.

Proof. If (1) holds, then $u^*uu^*u = u^*u$ and so $(uu^*u - u)^*(uu^*u - u) = u^*uu^*uu^*u - u^*uu^*u - u^*uu^*u + u^*u = 0$, so (2) holds, by the C^* -condition (as $a^*a = 0 \implies a = 0$ for $a \in A$). If (2) holds then multiply on the left by u^* to see that u^*u is idempotent; clearly u^*u is self-adjoint, and so (1) holds. Replacing u by u^* shows that (3) and (4) are equivalent. Finally, (2) and (3) are equivalent by taking adjoints. □

Definition 4.3. A *partial isometry* is $u \in A$ satisfying any of the above equivalent conditions. The projection u^*u is the *initial projection* and uu^* is the *final projection*.

Lemma 4.4. *For $U \in \mathcal{B}(H, K)$ we have that U is a partial isometry if and only if there is a closed subspace $H_0 \subseteq H$ such that U restricted to H_0 is an isometry, and U restricted to $H_0^\perp$ is 0. In this case, H_0 is the image of the initial projection, and $U(H_0)$ is the image of the final projection.*

Proof. Let U be a partial isometry, and let H_0 be the image of the initial projection U^*U . For $\xi = U^*U\xi \in H_0$, we have $\|U\xi\|^2 = (\xi|U^*U\xi) = (\xi|\xi) = \|\xi\|^2$, so conclude that U is an isometry on H_0 . By Lemma 4.1(2), we have $\ker U = \ker U^*U = H_0^\perp$.

Conversely, the restriction of U to H_0 is an isometry, and so $(U\xi|U\eta) = (\xi|\eta)$ for all $\xi, \eta \in H_0$, see Lemma 4.1(3). Let $\xi \in H$ and write $\xi = \xi_0 + \xi_1 \in H_0 \oplus H_0^\perp$, and similarly for η , so that $(\eta|U^*U\xi) = (U\eta|U\xi) = (U\eta_0|U\xi_0) = (\eta_0|\xi_0) = (\eta|\xi_0)$. Thus $U^*U\xi = \xi_0$, and so U^*U is the projection onto H_0 . Hence U is a partial isometry with initial space H_0 . Obviously $\text{Im}(UU^*) \subseteq \text{Im}(U)$, but $UU^*U = U$ so $UU^*\xi = \xi$ for any $\xi \in U(H_0)$, and we conclude that $\text{Im}(UU^*) = U(H_0)$. $\square$

For a partial isometry U on a Hilbert space, the *initial space* is the range of the projection U^*U and the *final space* is the range of UU^* .

We can now construct the polar decomposition. Given $T \in \mathcal{B}(H, K)$ form $|T| = (T^*T)^{1/2}$, and define

$$U: |T|(H) = \{|T|\xi : \xi \in H\} \rightarrow K; \quad U|T|\xi = T\xi.$$

For $\xi \in H$ we have that $\|T\xi\|^2 = (\xi|T^*T\xi) = \||T|\xi\|^2$ and so U is an isometry, and hence extends by continuity to the closure of $|T|(H)$. Set U to be 0 on the orthogonal complement of $|T|(H)$, so U is now defined on all of H . By construction, $U|T| = T$, and by Lemma 4.4, U is a partial isometry with initial space $\overline{\text{Im}}|T|$ and final space $\overline{\text{Im}}T$. By Lemma 4.1, $\overline{\text{Im}}|T| = (\ker |T|)^\perp = (\ker T)^\perp = \overline{\text{Im}}(T^*)$ and $\overline{\text{Im}}T = (\ker T^*)^\perp$. Also notice that $\ker U = (\text{Im } |T|)^\perp = \ker T$.

Proposition 4.5. *The polar decomposition is unique in the sense that if $T = VS$ with S positive and V a partial isometry with initial space $\overline{\text{Im}}S$ then $U = V$ and $S = |T|$. We have that $|T|$ belongs to the C^* -algebra generated by T^*T , and when $H = K$, U belongs to the von Neumann algebra generated by T .*

Proof. We have $T^*T = SV^*VS = S^2$ as V^*V is the projection onto $\overline{\text{Im}}S$, and so by uniqueness of positive square-roots, $|T| = S$. Then $U|T| = V|T|$ so U and V agree on $\overline{\text{Im}}|T|$ which is the initial space of both partial isomerities, and hence $U = V$.

By the continuous functional calculus, $|T| \in C^*(T^*T)$. When $H = K$, the von Neumann algebra generated by T is the bicommutant $\{T, T^*\}''$. Let S commute with T and T^* ; we need to show that $SU = US$. Let $\eta \in \ker U = \ker T$ so $TS\eta = ST\eta = 0$ so $S\eta \in \ker T$ so $US\eta = 0$. As S and $|T|$ commute, for $\xi \in H, \eta \in \ker U$, we have

$$SU(|T|\xi + \eta) = ST\xi = TS\xi = U|T|S\xi = US(|T|\xi + \eta).$$

As $|T|(H) + \ker U$ is dense in H , we conclude that $SU = US$, as required. $\square$

Suppose now that $T \in \mathcal{B}(H)$ is self-adjoint, so we can write $T = T_+ - T_-$ for some T_+, T_- positive with $T_+T_- = 0 = T_-T_+$. Let $H_\pm = \overline{\text{Im}}T_\pm$ so H_+ and H_- are mutually orthogonal as $(T_+\xi|T_-\eta) = (\xi|T_+T_-\eta) = 0$ for all $\xi, \eta \in H$. Let U be the operator which is 1 on H_+ , -1 on H_- and 0 on $(H_+ \oplus H_-)^\perp$. Then $U = U^*$ and U^*U is the projection onto $H_+ \oplus H_-$, so U is a partial isometry. As $\overline{\text{Im}}T = (\ker T)^\perp = (H_+ \oplus H_-)^\perp$ we see that U^*U is the projection onto $\overline{\text{Im}}T$. As $|T| = T_+ + T_-$ (by uniqueness, or functional calculus) we see that $U|T| = UT_+ + UT_- = T_+ - T_- = T$ and so by uniqueness, we have constructed the polar decomposition.

5 von Neumann regular elements

Again, we present some surely well-known results, with proofs, for applications to follow in the next section.

Definition 5.1. Let A be an algebra (or even just a ring). An element $x \in A$ is *von Neumann regular* if there is $y \in A$ with $xyx = x$.

Proposition 5.2. Let E be a Banach space and let $x \in \mathcal{B}(E)$. The following are equivalent:

1. x is von Neumann regular in $\mathcal{B}(E)$;
2. x has closed image and both $\ker(x)$ and $\operatorname{Im}(x)$ are complemented subspaces of E .

In this case, when $xyx = x$, we have that $xy = e$ is a projection onto $\operatorname{Im}(x)$ and $1 - yx = f$ is a projection onto $\ker(x)$. Conversely, given e and f projections onto $\operatorname{Im}(x)$, which is closed, and $\ker(x)$, respectively, we can choose y with $xyx = x$, $xy = e$ and with $yx = 1 - f$.

Proof. Suppose there is $y \in \mathcal{B}(E)$ with $xyx = x$. Then $(xy)^2 = xyxy = xy$ and $(yx)^2 = yxyx = yx$ so $e = xy$ and $f = 1 - yx$ are projections. We have $e(E) \subseteq x(E)$, but as $ex = xyx = x$ also $e(H) \supseteq x(E)$, so $x(E) = e(E)$ and in particular, x has closed, complemented image. As $xf = x - xyx = 0$ we have $f(E) \subseteq \ker(x)$, but if $x\xi = 0$ then $f\xi = \xi - yx\xi = \xi$ so we conclude that f is a projection onto $\ker(x)$.

Conversely, let x have closed image, with e a projection onto $x(E)$ and f a projection onto $\ker(x)$. Let x_0 be the restriction of x to a map $\operatorname{Im}(1 - f) \rightarrow \operatorname{Im}(e) = \operatorname{Im}(x)$. If $x_0\xi = 0$ then $(1 - f)\xi = \xi$ so $f\xi = 0$, but also $x\xi = 0$, so $f\xi = \xi$, so $\xi = 0$ and we conclude that x_0 is injective. Given any $\xi \in E$ we have that $xf\xi = 0$ so $x\xi = x(1 - f)\xi$ and hence $x_0(1 - f)\xi = x\xi$ and we conclude that x_0 is surjective. By the Open Mapping Theorem, x_0 is invertible. Set $y = x_0^{-1}e: E \rightarrow \operatorname{Im}(1 - f) \subseteq E$. Then $xy = xx_0^{-1}e = x_0x_0^{-1}e = e$. Given $\xi \in E$ set $\eta = (1 - f)\xi$ so as before, $x\xi = x\eta = x_0\eta$ as $\eta \in \operatorname{Im}(1 - f)$. Hence $yx\xi = x_0^{-1}ex\xi = x_0^{-1}x\xi = x_0^{-1}x_0\eta = \eta = (1 - f)\xi$. So $yx = 1 - f$. Finally, we have $xyx = ex = x$, as desired. $\square$

For the following, compare also Proposition 6.3 below.

Proposition 5.3. Let H be a Hilbert space, and let $x \in \mathcal{B}(H)$. Then x is von Neumann regular if and only if x has closed image. When x is von Neumann regular and self-adjoint, we can choose a self-adjoint y with $xyx = x$ and with $xy = yx$ a (self-adjoint) projection. Further, when x is positive, we can choose y positive.

Proof. In a Hilbert space, all closed subspaces are complemented, so x is von Neumann regular if and only if x has closed image. Now let x be self-adjoint with closed image. As $\operatorname{Im}(x) = (\ker x)^\perp$, Lemma 4.1, we can choose e to be the orthogonal projection onto $\operatorname{Im}(x)$, and then $f = 1 - e$ will be the orthogonal projection onto $\ker(x)$. Following the previous proof, let x_0 be the restriction of x to $\operatorname{Im}(1 - f) = \operatorname{Im}(e) = \operatorname{Im}(x)$, so $x_0: \operatorname{Im}(x) \rightarrow \operatorname{Im}(x)$ is invertible, and set $y = x_0^{-1}e$. Then $xy = e$, $yx = 1 - f = e$. As x is self-adjoint, also x_0 is self-adjoint, and so x_0^{-1} is self-adjoint. As $y = ey = ex_0^{-1}e$, we conclude that y is self-adjoint. The same argument shows that when x is positive, also y is positive. $\square$

6 Well-supported elements

This definition is given in [1, Definition II.3.2.8]. An element x in a C^* -algebra A is *well-supported* if $\sigma(x^*x) \setminus \{0\}$ is closed, that is, $\sigma(x^*x) \subseteq \{0\} \cup [\epsilon, \infty)$ for some $\epsilon > 0$. I haven't found an reference for elementary properties of such elements, so here give some elementary proofs, following Blackadar for the results.

As $\sigma(x^*x) \cup \{0\} = \sigma(xx^*) \cup \{0\}$, we see that x is well-supported if and only if x^* is. Let x be well-supported. Define $f: [0, \infty) \rightarrow [0, 1]$ by $f(0) = 0$ and $f(t) = 1$ for $t > 0$. Then f is continuous on $\sigma(x^*x)$ and so by the continuous functional calculus, $p = f(x^*x)$ exists, and p is a projection with $px^*x = x^*x = x^*xp$.

Lemma 6.1. *Let x be well-supported, and f be as above, and set $p = f(x^*x)$ and $q = f(xx^*)$. Then p is a projection, $xp = x$ and $xy = 0 \implies py = 0$. Also q is a projection, $qx = x$ and $yx = 0 \implies yq = 0$.*

Proof. As observed above, $px^*x = x^*x = x^*xp$ and so $(px^* - x^*)(xp - x) = px^*xp - x^*xp - px^*x + x^*x = 0$ hence $xp = x$. If $xy = 0$ then $x^*xyy^* = 0$ and $yy^*x^*x = y(xy)^*x = 0$ so working in the commutative C^* -algebra generated by x^*x, yy^* and 1, we see that $pyy^* = 0 = yy^*p$. Thus $pyy^*p = 0$ so $py = 0$.

As x^* is also well-supported, q is defined. We argue similarly: $(qx - x)(qx - x)^* = qxx^*q - xx^*q - qxx^* + xx^* = 0$ so $qx = x$. If $yx = 0$ then y^*y commutes with xx^* so $y^*yq = 0$ so $yq = 0$. $\square$

The next lemma says that for a well-supported element, its polar decomposition exists in the C^* -algebra.

Lemma 6.2. *Let $x \in A$ be well-supported, let $\pi: A \rightarrow \mathcal{B}(H)$ be a $*$ -homomorphism, and let $\pi(x) = u\pi(|x|)$ be the polar decomposition. Then $u \in \pi(A)$.*

Proof. Define $g: [0, \infty) \rightarrow [0, \infty)$ by $g(0) = 0$ and $g(t) = t^{-1/2}$, so again g is continuous on $\sigma(x^*x)$, and so we can set $v = xg(x^*x) \in A$. Then $g(x^*x)|x| = p$ by the functional calculus, where p is as above. So $v|x| = xp = x$. Also $v^*v = g(x^*x)x^*xg(x^*x) = p$ again by the functional calculus.

Let q be the projection onto $\overline{\text{Im}(\pi(|x|))} = \ker(\pi(|x|))^\perp = \ker(\pi(x))^\perp = \overline{\text{Im}(\pi(x^*))}$, and as $xp = x$ also $px^* = x^*$ so $p \geq q$. As $(1 - q)x^* = 0$, so $x(1 - q) = 0$, by Lemma 6.1, we have $p(1 - q) = 0$. So $p = pq = q$, and we conclude that v^*v is the projection onto $\overline{\text{Im}(\pi(|x|))}$. By uniqueness, Proposition 4.5, $\pi(x) = \pi(v)\pi(|x|)$ is the polar decomposition. $\square$

We now make links between well-supported elements and von Neumann regular elements, as considered in the previous section.

Proposition 6.3. *An element $x \in A$ is well-supported if and only if x is von Neumann regular meaning that there is $y \in A$ with $xyx = x$.*

Proof. Let x be well-supported. Again define $g(0) = 0, g(t) = t^{-1/2}$ for $t > 0$, so as in the proof of Lemma 6.2, with $u = xg(x^*x)$ we have that $x = u|x|$ is the polar decomposition. In particular, $uu^*u = u$. Set $y = g(x^*x)u^*$ so that $xyx = xg(x^*x)u^*u|x| = uu^*u|x| = u|x| = x$.

Conversely, let x be von Neumann regular. Represent A faithfully on some Hilbert space H , and let $x = u|x|$ be the polar decomposition, in $\mathcal{B}(H)$. Then $u^*x = |x|$ and so if we choose y with $xyx = x$ then $|x|yu|x| = |x|$, showing that $|x|$ is also von Neumann regular, in $\mathcal{B}(H)$. We now apply Proposition 5.3, which tells us that $|x|$ has closed image, and we can choose $y \geq 0$ in $\mathcal{B}(H)$ with $|x|y|x| = |x|$. Furthermore, $e = y|x| = |x|y$ is a projection onto $\text{Im}(|x|) = \ker(|x|)^\perp$.

Let $0 < \lambda \leq \|y\|^{-1}$. We show that $|x| - \lambda 1$ is bounded below. Let $\xi \in H$ so $|x|\xi = |x|e\xi = e|x|e\xi$. Hence, as e and $1 - e$ has orthogonal images,

$$\||x|\xi - \lambda\xi\|^2 = \||x|e\xi - \lambda e\xi - \lambda(1 - e)\xi\|^2 = \||x|e\xi - \lambda e\xi\|^2 + \lambda^2\|(1 - e)\xi\|^2.$$

We have $\|e\xi\| = \|y|x|\xi\| \leq \|y\|\||x|\xi\| = \|y\|\||x|y|x|\xi\| = \|y\|\||x|e\xi\|$ and so with $\eta = e\xi$ we see that $\||x|\eta\| \geq \|y\|^{-1}\|\eta\|$. So $\||x|\eta - \lambda\eta\| \geq \||x|\eta\| - \lambda\|\eta\| \geq (\|y\|^{-1} - \lambda)\|\eta\|$, and thus

$$\||x|\xi - \lambda\xi\|^2 \geq (\|y\|^{-1} - \lambda)^2\|e\xi\|^2 + \lambda^2\|(1 - e)\xi\|^2 \geq (\|y\|^{-1} - \lambda)^2\|\xi\|^2.$$

So $|x| - \lambda$ is bounded below and hence is an isomorphism onto its range. As we are working on a Hilbert space, there is $z \in \mathcal{B}(H)$ with $z(|x| - \lambda) = 1$, hence also $(|x| - \lambda)z^* = 1$. So $z = z^* = (|x| - \lambda)^{-1}$ and we conclude $\lambda \notin \sigma(|x|)$.

So $\sigma(|x|) \subseteq \{0\} \cup [c, \infty)$ for some $c > 0$ (we can take $c = \|y\|^{-1}$). Hence $\sigma(x^*x) = \sigma(|x|^2) \subseteq \{0\} \cup [c^2, \infty)$, and x is well-supported.¹ $\square$

Remark 6.4. The definition of being von Neumann regular might depend upon the algebra. However, notice that the proof of Proposition 6.3 shows that if $A \subseteq \mathcal{B}(H)$ and $x \in A$ is von Neumann regular in $\mathcal{B}(H)$, then there is $y \in A$ with $xyx = x$. Hence, for C^* -algebras, there is no dependence on the algebra.

According to Blackadar, sometimes well-supported elements are called *elements with closed range*, which is supported by the following, and the arguments in the previous proposition. First a lemma.

Lemma 6.5. *Let A be a C^* -algebra and let $x \in A$ be positive. Denote by $C^*(x) \subseteq A$ the C^* -algebra generated by x . For any $r > 0$ we have that $x^{1/2}$ is in the closure of $x^r C^*(x) \subseteq x^r A$.*

Proof. We use a functional calculus argument. Let $\epsilon > 0$ and define $f: [0, \infty) \rightarrow [0, \infty)$ by $f(t) = t\epsilon^{-1/2-r}$ for $t < \epsilon$, and $f(t) = t^{1/2-r}$ for $t \geq \epsilon$. Then f is continuous, and for $t \geq \epsilon$ we have $t^r f(t) = t^{1/2}$, while for $t < \epsilon$ we have $t^{1/2} - t^r f(t) = t^{1/2} - t^{1+r}\epsilon^{-1/2-r} = t^{1/2}(1 - (t/\epsilon)^{1/2+r}) \leq t^{1/2} \leq \epsilon^{1/2}$. Hence $\|x^r f(x) - x^{1/2}\| \leq \epsilon^{1/2}$. It follows that $x^{1/2}$ is in the closure of $x^r C^*(x)$. $\square$

Proposition 6.6. *Let $x \in A$. The following are equivalent:*

- (1) Ax is closed;
- (2) xA is closed;
- (3) x^*Ax is closed;
- (4) x is well-supported.

Proof. Let x be well-supported, so von Neumann regular, so there is y with $xyx = x$. Let (a_n) be a sequence in A with $a_n x \rightarrow a \in A$. Then $ayx = \lim_n a_n xyx = \lim_n a_n x = a$ and so $a = ayx \in Ax$. So (4) implies (1), and similarly (2). Similarly, if $x^* a_n x \rightarrow a \in A$ then $x^* y^* a y x = \lim_n x^* y^* x^* a_n x y x = \lim_n x^* a_n x = a$ so $a \in x^* Ax$, and so (4) implies (3).

Represent A faithfully on a Hilbert space H . Then Ax is closed in A if and only if Ax is closed in $\mathcal{B}(H)$. Let $x^* = v|x^*|$ be the polar decomposition, so $v^*x^* = |x^*|$. If Ax is closed in $\mathcal{B}(H)$ then for a sequence (a_n) in A with $a_n|x^*| \rightarrow a \in \mathcal{B}(H)$, we have $a_n x = a_n|x^*|v^* \rightarrow av^*$ so $av^* \in Ax$, say $av^* = bx$. Then $b|x^*| = bxv = av^*v = \lim_n a_n|x^*|v^*v = \lim_n a_n|x^*| = a$ and so $a \in A|x^*|$ and we conclude that $A|x^*|$ is closed. Conversely, suppose $A|x^*|$ is closed,

¹This argument could also be made, slightly more directly, by using the notion of the “approximate point spectrum”.

let $a_n x \rightarrow a$, so $a_n |x^*| = a_n x v \rightarrow a v \in A|x^*|$, say $av = b|x^*|$, so $bx = b|x^*|v^* = avv^* = \lim_n a_n x v v^* = \lim_n a_n |x^*|v^* v v^* = \lim_n a_n |x^*|v^* = \lim_n a_n x = a$ so $a \in Ax$ and we conclude Ax is closed. So we've shown that Ax is closed if and only if $A|x^*|$ is closed. Using the polar decomposition of x , one similarly shows that xA is closed if and only if $|x|A$ is closed. Finally, the same argument shows that x^*Ax is closed if and only if $|x^*|A|x^*|$ is closed.

Suppose that x is positive and xA is closed. Apply the lemma with $r = 1$ to conclude that $x^{1/2} \in \overline{xA} = xA$. Hence there is $a \in A$ with $xa = x^{1/2}$, and so $xaa^*x = x$ showing that x is von Neumann regular, and so well-supported. The same argument shows that if Ax is closed then x is well-supported. Thus if (1) holds, then $A|x^*|$ is closed, so $|x^*|$ is well-supported, but by definition, this means x^* is well-supported, so x is well-supported. So (4) holds. Similarly (2) implies (4).

Finally, suppose (3) holds, so $|x^*|A|x^*|$ is closed. Apply the lemma with $r = 2$ to conclude that $|x^*|^{1/2} \in \overline{|x^*|^2 C^*(|x^*|)} = \overline{|x^*|C^*(|x^*|)|x^*|} \subseteq \overline{|x^*|A|x^*|} = |x^*|A|x^*|$. So there is $a \in A$ with $|x^*|^{1/2} = |x^*|a|x^*|$ and so $|x^*|(a|x^*|^2 a^*)|x^*| = |x^*|$ showing that $|x^*|$ is von Neumann regular, hence well-supported. Again, this implies (4). $\square$

The above argument was inspired by [7, Theorem 8]. This paper [7] contains a little more about von Neumann regular elements, and following the citations will find similar papers.

Remark 6.7. We have tacitly worked with unital C^* -algebras, so we can smoothly apply the functional calculus. The continuous functional calculus is constructed by using Gelfand theory applied to a normal element $x \in A$. Then $C^*(1, x)$ is commutative and we find that it is isomorphic to $C(\sigma(x))$. Then the map sending x to the “coordinate function” $t \in C(\sigma(x))$ extends to polynomials, and by density, then to $C^*(1, x) \cong C(\sigma(x))$. The inverse map is the continuous functional calculus.

When A is non-unital, we embed A into A^+ the unitisation. By construction, if $f \in C(\sigma(x))$ can be approximated by polynomials with zero constant term, then $f(x) \in A$ not A^+ . By the locally compact space version of the Stone–Weierstrass theorem, [2, Corollary V.8.3], the collection of such f is exactly $C_0(\sigma(x) \setminus \{0\})$.

Hence $|x| = (x^*x)^{1/2} \in A$. All of the continuous functions we apply to well-supported elements vanish at 0 by construction, and so work in the non-unital case. The same applies to the function we used in the proof of Lemma 6.5.

6.1 Application to positive maps

Let A be a C^* -algebra (often a von Neumann algebra) and let $\varphi: A \rightarrow A$ be a (completely) positive map. We say that φ is *irreducible* if there is not a non-trivial projection p (so $p \neq 0, p \neq 1$) with $\varphi(p) \leq Mp$ for some $M > 0$. See [6, Proposition 1] for example, and references therein, for further details and motivation.

Lemma 6.8. *Let $v \in A$ be positive and invertible, and let $e \in A$ be a non-trivial projection. Then $x = vev$ is a well-supported element of A . The projection p associated to x , from Lemma 6.1, is non-trivial.*

Proof. There are a number of ways to show this: for example, it follows almost immediately from Proposition 6.6, as $Aev = Aev$ is closed if and only if Ae is closed, because v is invertible, and Ae is always closed as e is idempotent.

Let p be given by Lemma 6.1 applied to $x = vev$. Then $xp = x$ and $xy = 0 \implies py = 0$. As $x \neq 0$ clearly $p \neq 0$. If $p = 1$ then $xy = 0 \implies y = 0$, but let $y = v^{-1}(1 - e) \neq 0$, as $e \neq 1$, so see that $xy = vevv^{-1}(1 - e) = ve(1 - e) = 0$, contradiction. So p is non-trivial. $\square$

Proposition 6.9. *Let $v \in A$ be positive and invertible, let $\varphi: A \rightarrow A$ be positive, and set $\tilde{A}: A \rightarrow A; x \mapsto v^{-1}A(vxv)v^{-1}$, which is positive. Then A is irreducible if and only if $\tilde{A}$ is irreducible.*

Proof. Let $\tilde{A}$ be reducible, say $\tilde{A}(e) \leq Me$ for some non-trivial projection e and $M > 0$. By the lemma, $x = vev$ is positive and well-supported. Let $p \in A$ be the projection associated to x so p is non-trivial. As x is well-supported, $\sigma(x^2) \subseteq \{0\} \cup [c^2, C^2]$ for some $0 < c < C$, and so $\sigma(x) \subseteq \{0\} \cup [c, C]$. Let $f(0) = 0, f(t) = 1$ for $t > 0$, so by construction, $p = f(x)$. As $cf(t) \leq t \leq Cf(t)$ for $t \in \sigma(x)$, we see that $cp \leq x \leq Cp$. Then $\tilde{A}(e) \leq Me$ means $A(vev) \leq Mvev$, so $A(p) \leq c^{-1}A(x) \leq c^{-1}Mx \leq c^{-1}CMp$, in the first step using that A is positive. Thus A is reducible. Swapping the roles of v and v^{-1} shows that A reducible implies $\tilde{A}$ reducible. $\square$

This gives justification to the claim at the start of the proof of [3, Proposition 4.3]. There we work with a finite-dimensional C^* -algebra, for which it is obvious that all elements are well-supported. I thank Mateusz Wasilewski for correspondance which suggested the approach to proving Proposition 6.9 in the finite-dimensional case.

7 Inductive limits of Hilbert spaces

We use [11, Chapter XIV] as a reference, though this is standard material (and I suspect the main result of this section is in the literature, if I knew where to look). Let (H_n) be a sequence of Hilbert spaces with $\iota_n: H_n \rightarrow H_{n+1}$ isometries. Then (H_n, ι_n) is an *inductive sequence of Hilbert spaces*.

For $m \geq n$ define $\iota_{m,n} = \iota_{m-1} \circ \iota_{m-2} \circ \cdots \circ \iota_n$ an isometry $H_n \rightarrow H_m$, and set $\iota_{n,n} = 1_{H_n}$. As it is easy to forget the convention, we note this:

$$\iota_{m,n}: H_n \hookrightarrow H_m \quad (n \leq m).$$

Let H_∞ be the family of sequences $\xi = (\xi_n)$ where $\xi_n \in H_n$ for each n , and such that there is some m with $\xi_{n+1} = \iota_n(\xi_n)$ for $n \geq m$. That is, $\xi_n = \iota_{n,m}(\xi_m)$ for $n \geq m$. Notice that H_∞ is a vector space for the pointwise operations, and we may define a pre-inner-product by

$$(\xi|\eta) = \lim_n (\xi_n|\eta_n).$$

This is well-defined, as given ξ, η there is some m so that for $n \geq m$ we have $\xi_n = \iota_{n,m}(\xi_m)$ and $\eta_n = \iota_{n,m}(\eta_m)$, and so $(\xi_n|\eta_n) = (\iota_{n,m}(\xi_m)|\iota_{n,m}(\eta_m)) = (\xi_m|\eta_m)$ as $\iota_{n,m}$ is an isometry. Hence the sequence $(\xi_n|\eta_n)$ is eventually constant, and so the limit certainly exists. Notice that $(\xi|\xi) = 0$ if and only if $(\xi_n|\xi_n) = 0$ for sufficiently large n , and so we do not have an inner-product, and must quotient by the null space $\{\xi : (\xi|\xi) = 0\}$. The completion of the resulting space is $H = \varinjlim (H_n, \iota_n)$ a Hilbert space.

For each n we define

$$\iota_{\infty,n}: H_n \rightarrow H; \quad \xi \mapsto (0, 0, \dots, 0, \xi, \iota_{n+1,n}(\xi), \iota_{n+2,n}(\xi), \dots),$$

where the ξ occurs in the n th position. This is an isometry, and so we can regard $\iota_{\infty,n}$ as a map to H . We have that $\iota_{\infty,n+1} \circ \iota_n(\xi) = (0, \dots, 0, 0, \iota_{n+1,n}(\xi), \iota_{n+2,n}(\xi), \dots)$ which equals $\iota_{\infty,n}(\xi)$ in H (once we have quotiented H_∞ by the null space). Thus we have the commutative diagram:

$$\begin{array}{ccccccc} H_1 & \xrightarrow{\iota_1} & H_2 & \xrightarrow{\iota_2} & H_3 & \xrightarrow{\iota_3} & \cdots \\ & \searrow & \downarrow & \searrow & \downarrow & \searrow & \\ & & & & H & & \end{array}$$

$\iota_{\infty,1}$ $\iota_{\infty,3}$

The space H has the following universal property. Suppose K is a Hilbert space and for each n we have an isometry $u_n: H_n \rightarrow K$ with $u_{n+1} \circ \iota_n = u_n$, and such that the images of the u_n are dense in K . As $u_{n+1}(H_{n+1}) \supseteq u_n(H_n)$, we see that K is the closure of this increasing family of subspaces. For $\xi \in H_n$ define $U: u_n(\xi) \mapsto \iota_{\infty,n}(\xi) \in H$. As $u_n(\xi) = u_{n+1}\iota_n(\xi)$ and $\iota_{\infty,n+1}\iota_n(\xi) = \iota_{\infty,n}(\xi)$, we see that U is well-defined on the union of the subspaces $u_n(H_n)$. As all the maps are isometries, also U is an isometry, and so U extends to an isometry $U: K \rightarrow H$. The image of U contains each subspace $\iota_{\infty,n}(H_n)$, and as H is the union of all these subspaces, U is unitary.

7.1 Relation to the algebraic inductive limit

We recall the usual construction of the algebraic inductive limit. We consider the disjoint union of each space H_n and quotient by the equivalence relation that $\xi_i \in H_i$ is related to $\xi_j \in H_j$ if $\iota_{m,i}(\xi_i) = \iota_{m,j}(\xi_j)$ for some $m \geq i, j$. The relation is transitive as $\iota_{n,m} \circ \iota_{m,i} = \iota_{n,i}$ whenever $n \geq m \geq i$.

The vector space operations on each H_n drop to the equivalence classes. Given two classes $[\xi_i]$ and $[\xi_j]$ given $m \geq i, j$ we define $([\xi_i]|[\xi_j]) = (\iota_{m,i}(\xi_i)|\iota_{m,j}(\xi_j))_{H_m}$. As $\iota_{n,m}$ is an isometry for

$n \geq m$, we again see that this definition does not depend upon the choice of m . Furthermore, it's seen to be independent of the choice of vector giving the equivalence class.

The maps $\iota_{\infty,n}: H_n \rightarrow H_\infty$ respect the equivalence relation and give a bijection between the algebraic inductive limit and H_∞ , which is unitary for the inner-products we have defined. Thus we obtain the same construction.

7.2 As an inverse limit

The sequence (H_n, ι_n^*) is an inverse sequence, where a general inverse sequence is (H_n, q_n) where $q_n: H_{n+1} \rightarrow H_n$ is a coisometry for each n . Of course, every inverse sequence arises from an inductive sequence, and conversely, due to our requirement that each q_n be a coisometry.

We may define

$$q_{m,n} = q_n \circ q_{n+1} \circ \cdots \circ q_{m-1}: H_m \rightarrow H_n \quad (n \leq m).$$

The *algebraic inverse limit* is defined as all sequences $\xi = (\xi_n)$ where $\xi_n \in H_n$ for each n , and $q_{m,n}(\xi_m) = \xi_n$ for each $n \leq m$; equivalently, $\xi_n = q_n(\xi_{n+1})$ for each n . This space is a vector space for the pointwise operations. We define $\varprojlim (H_n, q_n)$ to be the subspace of the algebraic inverse limit consisting of all bounded sequences. For each n we define $q_{\infty,n}: \varprojlim (H_n, q_n) \rightarrow H_n$ to be the map $\xi \mapsto \xi_n$, that is, the projection onto the n th coordinate, which is a linear map.

Let $\xi = (\xi_n)$ with $\|\xi_n\| \leq K$ for each n . As each q_n is, in particular, a contraction, $\|\xi_n\| = \|q_n(\xi_{n+1})\| \leq \|\xi_{n+1}\|$ and so the sequence of norms $(\|\xi_n\|)$ is increasing, and bounded above, and so converges. For $n \leq m$, as $\iota_{m,n} = q_{m,n}^*: H_n \rightarrow H_m$ is an isometry, $q_{m,n}^* \circ q_{m,n}$ is an orthogonal projection of H_m onto the image of $q_{m,n}^*$, which is isometric with H_n . For a bounded sequence (ξ_n) we hence have

$$\|\xi_m\|^2 = \|q_{m,n}^* \circ q_{m,n}(\xi_m)\|^2 + \|\xi_m - q_{m,n}^* \circ q_{m,n}(\xi_m)\|^2 = \|\xi_n\|^2 + \|\xi_m - \iota_{m,n}(\xi_n)\|^2.$$

Thus $\|\xi_m - \iota_{m,n}(\xi_n)\|$ is small when $n \leq m$ and n is large. Given another bounded sequence η in the inverse limit, we have

$$\begin{aligned} (\xi_m | \eta_m) &= (\iota_{m,n}(\xi_n) + \xi_m - \iota_{m,n}(\xi_n) | \iota_{m,n}(\eta_n) + \eta_m - \iota_{m,n}(\eta_n)) \\ &= (\iota_{m,n}(\xi_n) | \iota_{m,n}(\eta_n)) + (\xi_m - \iota_{m,n}(\xi_n) | \iota_{m,n}(\eta_n)) + (\iota_{m,n}(\xi_n) | \eta_m - \iota_{m,n}(\eta_n)) \\ &\quad + (\xi_m - \iota_{m,n}(\xi_n) | \eta_m - \iota_{m,n}(\eta_n)) \\ &= (\xi_n | \eta_n) + (q_{m,n}(\xi_m) - \xi_n | \eta_n) + (\xi_n | q_{m,n}(\eta_m) - \eta_n) + (\xi_m - \iota_{m,n}(\xi_n) | \eta_m - \iota_{m,n}(\eta_n)) \\ &= (\xi_n | \eta_n) + (\xi_m - \iota_{m,n}(\xi_n) | \eta_m - \iota_{m,n}(\eta_n)), \end{aligned}$$

using that $\iota_{m,n}$ is an isometry, and that $q_{m,n}(\xi_m) = \xi_n$, and similarly for η . As the 2nd term is small, we see that the sequence $((\xi_n | \eta_n))$ is Cauchy and so converges. We may hence define an inner-product on the subspace of bounded sequences by $(\xi | \eta) = \lim_n (\xi_n | \eta_n)$. This is an inner-product, as $(\xi | \xi) = \lim_n \|\xi_n\|^2$ which in an increasing limit, and so equals 0 only when $\xi_n = 0$ for all n . So $\varprojlim (H_n, q_n)$ is an inner-product space.

Given $\xi = (\xi_n)$ in $\varprojlim (H_n, q_n)$, for large N , define $\eta = (\eta_n)$ by setting $\eta_n = \xi_n$ for $n \leq N$ and $\eta_n = \iota_{n,N}(\xi_N)$ for $n > N$. Then $q_n(\eta_{n+1}) = \eta_n$ for $n < N$, while for $n \geq N$ we have $q_n(\eta_{n+1}) = q_n q_{n+1,N}^*(\xi_N) = q_n q_n^* q_{n-1}^* \cdots q_N^*(\xi_N) = q_{n,N}^*(\xi_N) = \eta_n$. Thus η is in the inverse limit, is bounded, and for $n \geq N$ we have $\|\xi_n - \eta_n\| = \|\xi_n - \iota_{n,N}(\xi_N)\|$ is small. We conclude that the collection of such η is dense in $\varprojlim (H_n, q_n)$; notice also that $\eta \in H_\infty$.

Let $\xi = (\xi_n) \in H_\infty$, say $\xi_{n+1} = \iota_n(\xi_n)$ for $n \geq N$. Adjust ξ by setting $\xi_n = q_{N,n}(\xi_N)$ for $n \leq N$, and notice that this does not change the equivalence class that ξ defines in $\varprojlim (H_n, \iota_n)$. For $n < N$ we have $q_n(\xi_{n+1}) = q_n q_{N,n+1}(\xi_N) = q_{N,n}(\xi_N) = \xi_n$, while for $n \geq N$ we have $q_n(\xi_{n+1}) = \iota_n^* \iota_n(\xi_n) = \xi_n$. Hence $\xi \in \varprojlim (H_n, q_n)$ as of course (ξ_n) is bounded, and notice that

the norms of ξ in both spaces is the same. If $\xi' = (\xi'_n) \in H_\infty$ agrees with ξ in $\varinjlim(H_n, \iota_n)$ then $\xi_n = \xi'_n$ for sufficiently large n , and so we obtain the same sequence in $\varinjlim(H_n, q_n)$. We hence have an isometry $U: \varinjlim(H_n, q_n) \rightarrow \varinjlim(H_n, \iota_n)$. The previous paragraph shows that U has dense range, and so U is a unitary.

Thus the inverse and inductive limits agree; in particular, $\varinjlim(H_n, q_n)$ is a Hilbert space. Consider $q_{\infty, n}^*: H_N \rightarrow \varinjlim(H_n, q_n)$ which satisfies, for $\xi \in \varinjlim(H_n, q_n)$, $\eta_n \in H_n$,

$$\begin{aligned} (\xi | q_{\infty, n}^*(\eta_n)) &= (\xi_n | \eta_n) = \lim_n (q_{m, n}(\xi_m) | \eta_n) = \lim_m (\xi_m | \iota_{m, n}(\eta_n)) \\ &= (\xi | (q_{n, 1}(\eta_n), \dots, q_{n-1}(\eta_n), \eta_n, \iota_n(\eta_n), \iota_{n+1, n}(\eta_n), \dots)) \end{aligned}$$

This shows that $q_{\infty, n}^*(\eta_n) = (q_{n, 1}(\eta_n), \dots, q_{n-1}(\eta_n), \eta_n, \iota_n(\eta_n), \iota_{n+1, n}(\eta_n), \dots)$, where we can again check that this sequence is in $\varinjlim(H_n, q_n)$. As adjusting a finite number of elements doesn't change a sequence in $\varinjlim(H_n, q_n)$, we have shown that $U\iota_{\infty, n} = q_{\infty, n}U$. The inverse limit has the advantage that we obtain a concrete description of the entire Hilbert space, while the inductive limit only constructs a dense subspace. In particular, we have the following.

Proposition 7.1. *Let (H_n, ι_n) be an inductive sequence of Hilbert spaces. For each $\xi \in \varinjlim(H_n, \iota_n)$ there is a unique bounded sequence (ξ_n) with $\xi_n \in H_n$ for each n , with $\iota_n^*(\xi_{n+1}) = \xi_n$ for each n , and such that $\iota_{\infty, n}(\xi_n) \rightarrow \xi$.*

Conversely, if (ξ_n) is a sequence with $\xi_n \in H_n$ for each n , and “Cauchy” in the sense that for $\epsilon > 0$ there is N such that $\|\iota_{m, n}(\xi_n) - \xi_m\| < \epsilon$ for $m \geq n \geq N$, then there is $\xi \in \varinjlim(H_n, \iota_n)$ with $\iota_{\infty, n}(\xi_n) \rightarrow \xi$.

Proof. Given $\xi \in \varinjlim(H_n, \iota_n)$, let $(\xi_n) = U(\xi) \in \varinjlim(H_n, \iota_n)$. Then

$$\begin{aligned} U\iota_{\infty, n}(\xi_n) &= q_{\infty, n}^*U(\xi) = U(q_{n, 1}(\xi_n), \dots, q_{n, n-1}(\xi_n), \xi_n, \dots, \iota_{m, n}(\xi_n), \dots) \\ &= (q_{n, 1}(\xi_n), \dots, q_{n, n-1}(\xi_n), \xi_n, \dots, \iota_{m, n}(\xi_n), \dots), \end{aligned}$$

and so

$$\|U\iota_{\infty, n}(\xi_n) - U\xi\| = \lim_m \|\iota_{m, n}(\xi_n) - \xi_m\|,$$

which as observed before is small, if n is large. So $U\iota_{\infty, n}(\xi_n) \rightarrow U\xi$ and hence $\iota_{\infty, n}(\xi_n) \rightarrow \xi$.

For the second claim, we again work in $\varinjlim(H_n, \iota_n)$. For each n define $\eta^{(n)} = q_{\infty, n}^*(\xi_n) \in \varinjlim(H_n, \iota_n)$, so by definition, $\eta_k^{(n)} = \iota_{k, n}(\xi_n)$ for $k \geq n$. Given $\epsilon > 0$ select N as in the statement, so for $m \geq n \geq N$ and $k \geq m$ we have that

$$\|\eta_k^{(m)} - \eta_k^{(n)}\| = \|\iota_{k, m}(\xi_m) - \iota_{k, n}(\xi_n)\| \leq \|\iota_{k, m}(\xi_m) - \xi_k\| + \|\xi_k - \iota_{k, n}(\xi_n)\| < 2\epsilon.$$

Hence $\|\eta^{(m)} - \eta^{(n)}\| \leq 2\epsilon$ and so $(\eta^{(n)})$ is Cauchy, so convergent, in $\varinjlim(H_n, \iota_n)$. Thus $\iota_{\infty, n}(\xi_n)$ converges in $\varinjlim(H_n, \iota_n)$. $\square$

7.3 Application to infinite tensor products

Let H be a Hilbert space with unit vector $\xi_0 \in H$. Define $H_n = H^{\otimes n}$ and define connecting maps $\iota_n: H_n \rightarrow H_{n+1}; u \mapsto u \otimes \xi_0$. By definition, the infinite tensor product is $(H, \xi_0)^{\otimes \infty} = \varinjlim(H_n, \iota_n)$. The above proposition shows that we can regard this space as the limit points of sequences $u_n \in H^{\otimes n}$ which are Cauchy in the sense that $u_n \otimes \xi_0^{\otimes(m-n)} - u_m$ is small for $m \geq n$ large.

Suppose that $u_n = \xi_1 \otimes \dots \otimes \xi_n$ for each n . If $\lim_n \prod_{i \leq n} \|\xi_i\| = 0$ then $\|u_n\| \rightarrow 0$ and so $\|u_n \otimes \xi_0^{\otimes(m-n)} - u_m\| \leq \|u_n\| + \|u_m\|$ is small when n, m are large. Then of course the limit

in $(H, \xi_0)^{\otimes \infty}$ is 0. Similarly, if $\lim_n \prod_{i \leq n} \|\xi_i\| = \infty$ then $\iota_{\infty, n}(u_n)$ is unbounded, and so cannot converge.

So we assume that $0 < \lim_n \prod_{i \leq n} \|\xi_i\| < \infty$. For $m \geq n$ we have

$$\begin{aligned} & \|\xi_1 \otimes \cdots \otimes \xi_n \otimes \xi_0 \otimes \cdots \otimes \xi_0 - \xi_1 \otimes \cdots \otimes \xi_m\|^2 \\ &= \|\xi_1 \otimes \cdots \otimes \xi_n \otimes (\xi_0 \otimes \cdots \otimes \xi_0 - \xi_{n+1} \otimes \cdots \otimes \xi_m)\|^2 \\ &= \prod_{i=1}^n \|\xi_i\|^2 \left(1 + \prod_{i=n+1}^m \|\xi_i\|^2 - 2\Re \prod_{i=n+1}^m (\xi_0 | \xi_i) \right). \end{aligned}$$

The first term is bounded away from 0 and ∞ , so we look at the term in brackets, which must be small for large n .

We now make some remarks about [11, Lemma XIV.1.7], in particular giving a counterexample to one direction of this claim. We start by looking at some elementary analysis results about infinite products, for which we follow the lecture notes [9], although we allow an infinite product to converge to 0.

Lemma 7.2. *Let (z_n) be a sequence in $\mathbb{C}$. The following are equivalent:*

1. $\lim_n \prod_{i \leq n} z_i$ exists and is non-zero;
2. no z_n is zero, and for each $\epsilon > 0$ there is N such that for each $m \geq n \geq N$ we have $|1 - \prod_{i=n}^m z_i| < \epsilon$.

Proof. Suppose $\lim_n \prod_{i \leq n} z_i \neq 0$, so no z_i is 0, and setting $p_n = \prod_{i \leq n} z_i$, there is $\delta > 0$ so that $|p_n| > \delta$ for all n . There is N so that $|p_n - p_m| < \epsilon\delta$ for each $m \geq n \geq N$. Then $|1 - \prod_{i=n+1}^m z_i| = |1 - p_m/p_n| = |p_n|^{-1} |p_n - p_m| < \epsilon$.

For the converse, given $\epsilon > 0$ select N . Then $|1 - z_n| < \epsilon$ for each $n \geq N$ so $1 - \epsilon < |z_n| < 1 + \epsilon$. Set $q_m = z_N z_{N+1} \cdots z_m$ for $m \geq N$, so we also have $|1 - q_m| < \epsilon$. In particular, we may suppose that $\epsilon < 1/2$ so that $1/2 < |q_m| < 3/2$ for all $m \geq N$. Increasing N if necessary, we may suppose that for $m > n \geq N$,

$$\left| \frac{q_m}{q_n} - 1 \right| = \left| \prod_{i=n+1}^m z_i - 1 \right| < \frac{2}{3}\epsilon.$$

Then $|q_m - q_n| = |q_n| |q_m/q_n - 1| < \frac{2}{3}\epsilon |q_n| < \frac{3}{2} \frac{2}{3}\epsilon = \epsilon$. Hence $(q_n)_{n \geq N}$ is Cauchy and so converges. As $\prod_{i=1}^n z_i = q_n \prod_{i=1}^{N-1} z_i$, also the infinite product converges. As $z_i \neq 0$ for each i , and as $|q_n| > 1/2$, we also see that the limit is non-zero. $\square$

Proposition 7.3. *Let (a_n) be a sequence of positive reals. Then $\lim_n \prod_{i \leq n} (1 + a_i)$ converges if and only if $\sum_n a_n$ converges.*

Proof. By hypothesis, both series are increasing, and so converge if and only if they are bounded above. As $x > 0$ implies $1 + x < e^x$ we see that $\sum_{i \leq n} a_i \leq \prod_{i \leq n} (1 + a_i) \leq \exp(\sum_{i \leq n} a_i)$ and the result follows. $\square$

Note of course that if $\lim_n \prod_{i \leq n} (1 + a_i)$ converges then its limit is > 1 . We next come to a notion of “absolute convergence”.

Corollary 7.4. *Let (a_n) be a sequence in $\mathbb{C}$ such that $\sum_n |a_n|$ converges. Then $\lim_n \prod_{i \leq n} (1 + a_i)$ converges (and is non-zero if $a_i \neq -1$ for all i).*

Proof. Note that

$$\begin{aligned} & |(1 + a_n)(1 + a_{n+1}) \cdots (1 + a_m) - 1| = |a_n + \cdots + a_m + \sum_{n \leq i < j \leq m} a_i a_j + \cdots| \\ & \leq |a_n| + \cdots + |a_m| + \sum_{n \leq i < j \leq m} |a_i a_j| + \cdots = (1 + |a_n|)(1 + |a_{n+1}|)(1 + |a_m|) - 1, \end{aligned}$$

and so Lemma 7.2 shows that $\lim_n \prod_{i \leq n} (1 + |a_n|)$ converges implies that $\lim_n \prod_{i \leq n} (1 + a_n)$ converges (and is non-zero if $a_i \neq -1$ for all i). If $\sum_n |a_n|$ converges, then by Proposition 7.3, $\lim_n \prod_{i \leq n} (1 + |a_i|)$ converges, and the result follows. $\square$

We now come to one direction of [11, Lemma XIV.1.7].

Proposition 7.5. *Let (ξ_n) be a sequence in H , and set $u_n = \xi_1 \otimes \cdots \otimes \xi_n$ for each n . If $\lim_n \|u_n\| = \lim_n \prod_{i \leq n} \|\xi_i\|$ converges and is non-zero, and $\sum_n |1 - (\xi_n | \xi_0)| < \infty$, then (u_n) converges in the infinite tensor product.*

Proof. From the discussion above, given $\epsilon > 0$ we seek N so that if $m \geq n \geq N$ we have

$$1 + \prod_{i=n+1}^m \|\xi_i\|^2 - 2\Re \prod_{i=n+1}^m (\xi_0 | \xi_i) < \epsilon.$$

Set $a_n = 1 - (\xi_0 | \xi_n)$ so by hypothesis, $\sum_n |a_n| < \infty$ and so Corollary 7.4 tells us that $\lim_n \prod_{i \leq n} (1 + a_n) = \lim_n \prod_{i \leq n} (\xi_0 | \xi_i)$ converges and is non-zero. By Lemma 7.2 there is N so that for $m \geq n \geq N$ we have both

$$\left| 1 - \sum_{i=n+1}^m (\xi_0 | \xi_i) \right| < \epsilon, \quad \left| 1 - \sum_{i=n+1}^m \|\xi_i\|^2 \right| < \epsilon.$$

For $z \in \mathbb{C}$, as $|2 - 2\Re z| = |2 - z - \bar{z}| \leq |1 - z| + |1 - \bar{z}|$, we see that $|2 - 2\Re \sum_{i=n+1}^m (\xi_0 | \xi_i)| < 2\epsilon$, and so

$$\left| 1 + \prod_{i=n+1}^m \|\xi_i\|^2 - 2\Re \prod_{i=n+1}^m (\xi_0 | \xi_i) \right| = \left| \sum_{i=n+1}^m \|\xi_i\|^2 - 1 + 2 - 2\Re \prod_{i=n+1}^m (\xi_0 | \xi_i) \right| < 3\epsilon,$$

as we want. $\square$

The converse does not seem to hold, as we now show. We continue to follow [9].

Proposition 7.6. *Let $a_n \geq 0$ for each n . If $\sum_n a_n$ converges then $\lim_n \prod_{i \leq n} (1 - a_i)$ converges. If $\lim_n \prod_{i \leq n} (1 - a_i)$ converges then either the limit is 0 or $\sum_n a_n$ converges.*

Proof. If $\sum_n a_n$ converges then Corollary 7.4 tells us that $\lim_n \prod_{i \leq n} (1 - a_i)$ converges.

Conversely, suppose $\lim_n \prod_{i \leq n} (1 - a_i)$ converges and is non-zero, so by Lemma 7.2, in particular, $1 - (1 - a_i) = a_i \rightarrow 0$, so there is N such that $a_n < 1$ for $n \geq N$. Towards a contraction, suppose that $\sum_n a_n$ diverges, so also $\lim_n \prod_{i \leq n} (1 + a_i)$ diverges. As $(1 - a_i)(1 + a_i) = 1 - a_i^2 \leq 1$ for $i \geq N$, for $m \geq n \geq N$ we have

$$\prod_{i=n}^m (1 - a_i) \prod_{i=n}^m (1 + a_i) = \prod_{i=n}^m (1 - a_i^2) \leq 1 \quad \implies \quad \prod_{i=n}^m (1 - a_i) \leq \left(\prod_{i=n}^m (1 + a_i) \right)^{-1}.$$

Thus $\prod_{i=n}^m (1 - a_i)$ is arbitrarily small, but by Lemma 7.2, is also arbitrarily close to 1, giving the claimed contraction. $\square$

We now come to our counter-example. Let

$$(a_n) = \left(1 - \frac{1}{\sqrt{2}}, 1 + \frac{1}{2} + \frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{3}}, 1 + \frac{1}{3} + \frac{1}{\sqrt{3}}, \dots\right).$$

Then $\sum_n |1 - a_n| \geq \sum_n n^{-1/2} = \infty$. However, we claim that $\lim_n \prod_{i \leq n} a_i$ converges, and is non-zero. We have

$$\left(1 - \frac{1}{\sqrt{k}}\right)\left(1 + \frac{1}{k} + \frac{1}{\sqrt{k}}\right) = \frac{1}{k}\left(1 - \frac{1}{\sqrt{k}}\right) + 1 - \frac{1}{k} = 1 - \frac{1}{k\sqrt{k}}.$$

By Proposition 7.6, $\lim_n \prod_{i \leq n} \left(1 - \frac{1}{i\sqrt{i}}\right)$ converges and is non-zero, and so by Lemma 7.2, for $\epsilon > 0$ there is N so that if $m \geq n \geq N$ we have

$$\left|1 - \prod_{i=2n-1}^{2m} a_i\right| < \epsilon.$$

Then

$$\sum_{i=2n}^{2m} a_i = \left(1 - \frac{1}{\sqrt{n+1}}\right)^{-1} \prod_{i=2n-1}^{2m} a_i, \quad \sum_{i=2n-1}^{2m+1} a_i = \left(1 + \frac{1}{\sqrt{m}} + \frac{1}{\sqrt{m+1}}\right) \prod_{i=2n-1}^{2m} a_i,$$

which are then both within 2ϵ of 1, if N is large enough, and similarly for the final case of $\sum_{i=2n}^{2m+1} a_i$. By Lemma 7.2 applied in the other direction, we conclude that $\lim_n \prod_{i \leq n} a_i$ converges and is non-zero.

With our Hilbert space example from the previous section, set $\xi_n = a_n \xi_0$ for each n . Then

$$\xi_1 \otimes \dots \otimes \xi_n = a_1 a_2 \dots a_n \xi_0 \otimes \dots \otimes \xi_0,$$

and so for $m \geq n$,

$$\begin{aligned} & \|\xi_1 \otimes \dots \otimes \xi_n \otimes \xi_0 \otimes \dots \otimes \xi_0 - \xi_1 \otimes \dots \otimes \xi_m\| \\ &= \left| \prod_{i=1}^n a_i - \prod_{i=1}^m a_i \right| = \left| \prod_{i=1}^n a_i \right| \left| 1 - \prod_{i=n+1}^m a_i \right|. \end{aligned}$$

As $\lim_n \prod_{i \leq n} a_i$ converges, this is arbitrarily small if n is large enough. So the sequence $(\xi_1 \otimes \dots \otimes \xi_n)$ converges in the infinite tensor product. However, $\sum_n |1 - (\xi_n | \xi_0)| = \sum_n |1 - a_n|$ diverges, contrary to [11, Lemma XIV.1.7].

Finally, we remark that we can use the relationsation of $(H, \xi_0)^{\otimes \infty}$ as the inverse limit to write down infinite tensors directly. We have that

$$\iota_n^*(\xi_1 \otimes \dots \otimes \xi_{n+1}) = (\xi_0 | \xi_{n+1}) \xi_1 \otimes \dots \otimes \xi_n.$$

Hence $(u_n) = (\xi_1 \otimes \dots \otimes \xi_n)$ is in the inverse limit, we might write $\xi_1 \otimes \xi_2 \otimes \dots \in (H, \xi_0)^{\otimes \infty}$, if and only if $(\xi_0 | \xi_i) = 1$ for all $i > 1$, and $\lim_n \prod_{i \leq n} \|\xi_i\| < \infty$.

The boundedness condition implies that $\|\xi_i\| \rightarrow 1$, and so $\xi_i = \xi_0 + \eta_i$ say, where $\eta_i \in \{\xi_0\}^\perp$ and $\|\eta_i\| \rightarrow 0$. Then $\|\xi_i\|^2 = 1 + \|\eta_i\|^2$, and so $\lim_n \prod_{i \leq n} \|\xi_i\| < \infty$ if and only if $\lim_n \prod_{i \leq n} (1 + \|\eta_i\|^2) < \infty$, which by Proposition 7.3 is equivalent to $\sum_i \|\eta_i\|^2 < \infty$.

8 Some combinatorics related to Catalan numbers

See https://en.wikipedia.org/wiki/Catalan_number for the basics. Consider the different ways of associating n applications of a binary operation. For example, when $n = 4$ we have

$$((ab)c)d, \quad (a(bc))d, \quad (ab)(cd), \quad a((bc)d), \quad a(b(cd)).$$

The number of ways to do this is the Catalan number C_{n-1} .

How can we decide if a “bracketting” is valid? After some thought, I can up with the following recursive definition. A string of variables and brackets X is valid exactly when X is of the form xy or $x(Y)$ or $(Y)x$ or $(Y)(Z)$ where x, y are variables, and Y, Z are valid, non-trivial brackettings. Here non-trivial means not empty or just a single variable. For example $X = ((ab)c)d$ is valid, as $X = (Y)d$ where $Y = (ab)c$ is valid, because $Y = (Z)c$ and $Z = ab$ is valid. To see why this is true, notice that a valid bracketting must give us a way to repeatedly apply the binary operations (here written multiplicatively) in an unambiguous way. The final step in evaluating X must hence be two operands: either single variables, or bracketted expressions. Furthermore, any bracketted expressions must have been valid, or else we would not have been able to evaluate them.

Proposition 8.1. *When the binary operation is associative, $a(bc) = (ab)c$, any way of associating n applications of the binary product gives the same answer.*

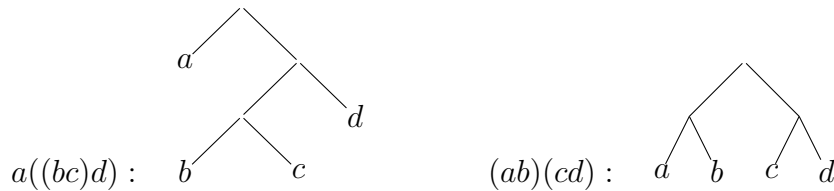
Proof. We follow the proof Dubuque, [5]. Say that a choice of parentheses is in (*right associated*) *normal form* when it's of the form $a_1(a_2 \cdots (a_{n-1}a_n) \cdots)$, for example, $a(b(cd))$. We will show, only using the associativity assumption, that any bracketing of n operands can be transformed to one that is in normal form. The proof is by (total) induction on $n = \ell(X)$ where X is the expression which has n operands.

The case $\ell(X) = 1$ is the base case, trivially true. As discussed above, when $\ell(X) > 1$ we must have that $X = YZ$ where Y and Z are either single variables, or expressions which start with “(” and end with “)”. For example, $X = a((bc)d)$ splits as $Y = a$ and $Z = ((bc)d)$. For a further example, $X = (ab)(cd)$ splits as $Y = (ab)$ and $Z = (cd)$. There are then two cases:

- Y is a single variable (namely a_1). As $\ell(Z) < \ell(X)$ by induction Z is equivalent to $(a_2(a_3 \cdots))$ normal form (or $Z = a_2$ is a single variable with no brackets), and then $X = a_1(a_2(\cdots))$ (or $X = a_1a_2$) is also in normal form.
- $\ell(Y) > 1$, as we are not allowed to write “ (a_1) ”. In particular Y starts and ends with a bracket. As $\ell(Y) < \ell(X)$ by induction Y is equivalent to $(a_1(a_2 \cdots (a_{k-1}a_k) \cdots))$ normal form and so $X = (a_1(a_2 \cdots (a_{k-1}a_k) \cdots))Z = (a_1Y')Z$ say. By associativity, also $X = a_1(Y'Z)$. A corner-case is when $X = (a_1a_2)$, but then $X = a_1(a_2Z)$ is valid (either $Z = a_3$ or $Z = (Z')$). This is now in the form of the first case.

□

Any way to fully bracket n operands can be visualised as a binary tree. For example:



Given a binary tree with n leaves, we assume the tree is drawn as above, or at least, that at any non-leaf node, a “left” and “right” child node are specified. Thus any tree can be split at the root into the left and right sub-tree, unless the tree consists of just a leaf. When we just have a left (the base case for the induction / recursion) we obtain the expression “ a_1 ”. Two expressions L and R are combined as follows: relabel the operands of R to start with $a_{\ell(L)+1}$, and then add parentheses if the expression has length greater than one. For example, $L = a_1, R = a_1$ becomes a_1a_2 , while $L = a_1, R = (a_1a_2)a_3$ becomes $a_1((a_2a_3)a_4)$. This recursive algorithm allows us to associate an expression to a tree.

Starting with a bracketted expression and producing a binary tree is similar: we use the discussion in the proof that X splits as YZ with Y and Z either being single variables (so a leaf child node) or start and end with brackets: recursively finding the tree associated to the expression inside the brackets gives a non-leaf child node.

In this way we obtain a bijection between binary trees with n leaves and valid bracket expressions involving n variables.

A *Dyck word* is a string of n x s and n y s so that no initial substring ever contains more y s than x s. E.g. $xyxy$ or $xyxy$ are the two valid words with $n = 2$. If we think of x as “(” and y as “)” then such words are exactly all the strings of matched pairs of brackets: $(())$ or $()()$ in the $n = 2$ case. There is a (non-obvious) bijection between such words and binary trees with $n + 1$ leaves. We follow [12].

Any Dyck word X of length $2(n + 1)$ (thought of as a bracket match) starts with “(”, so find the matching “)” and let Y be the length $2i$ string between these two brackets, and Z the remainder, of length $2(n - i)$, so $X = (Y)Z$. Notice $i = 0$ and $i = n$ are allowed. This gives a recursive formula for the number of Dyck words,

$$C_{n+1} = \sum_{i=0}^n C_i C_{n-i},$$

which shows that we obtain the Catalan numbers.

We seek a similar decomposition of a binary tree with $n + 1$ leaves. An *internal node* of a binary tree is a non-leaf node: in the bijection above an internal node corresponds to applying the binary operation, and so we see immediately that there are n internal nodes. (The tree with 1 leaf is just the tree with a root and nothing else, and the tree with 2 leaves has a root and two children.) Given any binary tree with n internal nodes, delete the root to obtain two subtrees, say with i and $n - i$ internal nodes. Hence we arrive at the same recursion.

This gives the hint as to how to obtain a bijection: the tree with just a root corresponds to the empty Dyck word, and the tree with a root and two children corresponds to the Dyck word $()$. Then, recursively:

1. $a(bc)$ splits as a and bc corresponding to $\emptyset$, $()$ which becomes the word $()()$;
2. $(ab)c$ splits as ab and c corresponding to $()$ and $\emptyset$ which becomes $(())$.

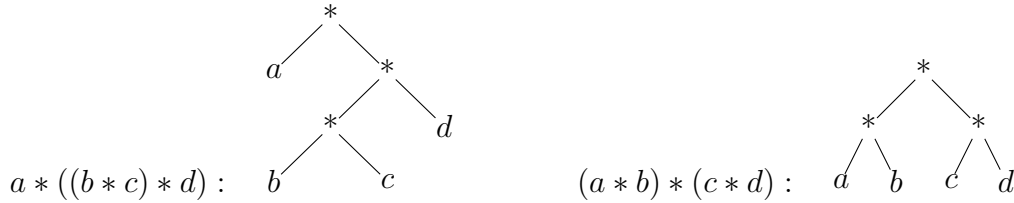
We now consider the $n = 3$ case:

1. $((ab)c)d \rightarrow (ab)c, d$ becomes $((()))$;
2. $(a(bc))d \rightarrow a(bc), d$ becomes $((()))$;
3. $(ab)(cd) \rightarrow ab, cd$ becomes $(())()$;
4. $a((bc)d) \rightarrow a, (bc)d$ becomes $()()$;
5. $a(b(cd)) \rightarrow a, b(cd)$ becomes $()()$.

Notice the pattern: the first Dyck word is enclosed in further brackets, while the second Dyck word is simply appended.

As described in [12] there is another way to see this: start with a bracketting of $n + 1$ variables, add a symbol “*” when we apply the binary product, delete the variables, add outer brackets. E.g. $((ab)c)d \rightarrow ((a * b) * c) * d \rightarrow (((*) * *) *)$ or $(ab)(cd) \rightarrow (a * b) * (c * d) \rightarrow ((*) * (*))$. Now delete the closing brackets, and then replace the “*”s by a closing bracket. So $(((*) * *) *) \rightarrow ((()))$ and $((*) * (*)) \rightarrow (())()$.

The operation of adding a “*” is the same as highlighting the internal nodes of the tree:



Deleting the root node and splitting the tree corresponds to deleting the “top level” * to obtain two sub-expressions. Adding outer brackets, then deleting variables, closing brackets and changing the * to a closing bracket means that the first (left) sub-expression is surrounded by brackets (the added opening bracket, and the top level * which becomes a closing bracket) while the second (right) sub-expression has nothing happen to it. This is now seen to be the same as the recursive procedure.

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